

Research Proposal: Route-Based Accessibility and the San Tin Technopole Capitalization Gradient

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1 Research Question and Motivation

This proposal asks whether residential estates in northern Hong Kong with greater accessibility to the San Tin Technopole (STT) experienced different price changes after the Draft Outline Zoning Plan was gazetted on 8 March 2024, and whether Google driving time captures that response more sharply than Euclidean distance. Treatment intensity is estate accessibility to the nearest internal STT transport node, measured either by negative log Google driving time or negative log Euclidean distance. The outcome is log transaction price per square foot, and the unit of observation is a sale transaction.

The question matters for policy because STT is a core project of the Northern Metropolis strategy. Its expected price effect is ambiguous because the gazettal bundled two opposing information shocks: long-run development and accessibility gains, and short-run local exposure costs from the associated land-resumption proposal covering 1,776 private lots and about 171 hectares. The setting is also methodologically useful: the Shenzhen River and sparse road network create large detours, driving a wedge between straight-line and route-based access.

I estimate the short-run announcement gradient of this mixed shock and test whether the inferred capitalization pattern depends on how accessibility is measured.

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2 Institutional Background

The STT is a 627-hectare innovation-and-technology cluster planned adjacent to the Shenzhen River and the Lok Ma Chau Loop. The Draft OZP No. S/STT/1 was gazetted under Section 5 of the Town Planning Ordinance on 8 March 2024 and placed on public exhibition through a same-day government press release.¹ The plan sets out the statutory zonings, road reserves, and four future railway-and-transport interchanges that structure the technopole's internal mobility.

The 8 March release also announced the San Tin land-resumption proposal, covering 1,776 private lots and about 171 hectares, and described the future treatment of brown-field operations in the affected fringe. A later milestone, the approved OZP No. S/STT/2 of 20 September 2024, confirmed the plan but added no first-order information on resumption boundaries or construction timetables. I use this sequence to define the main shock at the draft-plan gazettal and to implement an alternative shock-date check at the approval date (Section 7).

3 Related Literature and Contribution

I build on three strands of empirical urban economics. First, Rosen (1974) provides the hedonic framework for interpreting price gradients in characteristics space. I use a DiD-style hedonic equation to estimate how the accessibility gradient changes at the announcement date.

A second strand studies how transport investments are capitalized into housing prices. In Hong Kong, Jayantha et al. (2015) show that anticipated transport improvements are capitalized before completion, while Bao and Mok (2020) find spatially uneven capitalization from the Express Rail Link. Rojas (2024) distinguish direct accessibility

¹Hong Kong Government, "New San Tin Technopole Outline Zoning Plan gazetted and two approved Outline Zoning Plans amended."

effects from indirect effects generated by complementary non-transport investments around new stations. The STT setting differs because the gazettal bundles long-run accessibility gains with an immediate land-resumption announcement, so the short-run sign is not determined in advance.

A third strand concerns how accessibility is measured. Apparicio et al. (2008) and Buczkowska et al. (2019) show that accessibility measures can differ materially across metrics when straight-line distance is a poor proxy for actual travel. The Shenzhen River and sparse road network make STT a useful setting for comparing route-based and Euclidean access. I frame the design as a continuous-treatment DiD-style exercise informed by Callaway et al. (2024), and report IQR effects and standardized coefficients to compare differently scaled treatments, following Nieminen (2022) and Oka (2021). My contribution is to bring these strands together and show how much the accessibility metric changes the substantive conclusion.

4 Data and Variables

I collect transaction records from the public listings of Centaline Property Agency Limited (*Centanet*).² From these listings, I scrape the maximum publicly accessible window of 283,173 records. I then obtain estate coordinates from the Google Geocoding API³ and estate-to-STT travel times from the Google Routes API.⁴ Because the earliest sample dates lie within 3 years of the query and the road network around STT has been broadly stable, I treat contemporaneous Google driving times as a defensible proxy for the route accessibility buyers would have inferred at the time of transaction. The full pipeline and replication repository are documented in Appendix A.

The unit of observation is a single transaction, indexed by (i, j, t) , where i denotes the estate, j the transaction within an estate-month, and t the calendar month. I restrict the

²Centaline Property Agency Limited, [public transaction listings](#).

³Google Maps Platform, [Geocoding API documentation](#).

⁴Google Maps Platform, [Routes API documentation](#).

data to sale transactions with a non-empty estate name and to the 6 *web_scope* areas nearest to STT: Yuen Long; Sheung Shui, Fanling, and Kwu Tung; Tin Shui Wai; Kam Tin, Palm Springs/Royal Palms, and The Vineyard; Tuen Mun; and Tai Po. This leaves a working sample of 34,495 transactions linked to 720 unique estate-level geocode queries. For the main analysis, I further restrict dates to a symmetric 1-year window around the gazettal, [2023-03-08, 2025-03-07], yielding 16,733 transactions over 505 estates and 24 months.

Let T_i^G denote the shortest Google driving time (in minutes) from estate i to the 4 STT internal interchanges, and let D_i^E denote the shortest non-network distance (in kilometres), computed as a great-circle distance from coordinates. I refer to the latter as the Euclidean benchmark for brevity and report the calculation in Appendix A. Each transaction has a calendar date τ_{ijt} which falls inside month t . The accessibility variables and the transaction-level post indicator are

$$A_i^G = -\log(T_i^G + 1), \quad A_i^E = -\log(D_i^E + 1), \quad \text{post}_{ijt} = \mathbf{1}\{\tau_{ijt} \geq 2024-03-08\}. \quad (1)$$

The transaction-level continuous-treatment intensity is $D_{ijt}^m = A_i^m \cdot \text{post}_{ijt}$ for metric $m \in \{G, E\}$. The outcome is $Y_{ijt} = \log(\text{price per sq. ft.})$. The control vector X_{ijt} contains log saleable area, a floor-height bin, building age, and a first- or second-hand indicator. Estate and month fixed effects are α_i and δ_t , with δ_t defined at the calendar-month level. Standard errors are clustered by estate throughout.

5 Econometric Strategy and Identification

I estimate a continuous-treatment DiD-style hedonic regression in the sense of Callaway et al. (2024). All estates face the same announcement on the same day, but treatment intensity varies with accessibility. My transaction-level baseline specification is

$$Y_{ijt} = \beta (A_i^m \cdot \text{post}_{ijt}) + X_{ijt}'\gamma + \alpha_i + \delta_t + \varepsilon_{ijt}, \quad m \in \{G, E\}, \quad (2)$$

where estate fixed effects α_i absorb the level term A_i^m , so identification of β comes from within-estate changes between pre and post transactions interacted with the cross-sectional accessibility gradient. I estimate equation (2) 4 times, with D^G and D^E , each with and without the controls X_{ijt} .

Under the standard continuous-treatment exogeneity condition

$$\mathbb{E} \left[\varepsilon_{ijt} \mid A_i^m \cdot \text{post}_{ijt}, X_{ijt}, \alpha_i, \delta_t \right] = 0, \quad (3)$$

β measures the average change in log price per square foot associated with a one-unit increase in accessibility after the gazettal, the continuous-treatment analogue of a marginal ATT. The key identifying assumption is parallel trends: absent the gazettal, the price-growth gap between more and less accessible estates would have remained stable over time. Estate fixed effects absorb time-invariant differences across estates, while month fixed effects absorb shocks common to all estates in a given calendar month.

I probe the main threats to this interpretation in Section 7. Web-scope-by-post interactions test whether area-specific post-period movements explain the estimated gradient; first- versus second-hand controls test whether transaction composition drives the result; and dropping the nearest estates checks whether the estimate is concentrated among a small set of near-STT outliers. Separately, I use the Euclidean specification as a within-paper benchmark for assessing whether the substantive conclusion depends on how accessibility is measured.

Because the short-run sign is theoretically ambiguous, I use 2-sided tests,

$$\mathbb{H}_0 : \beta = 0, \quad \mathbb{H}_1 : \beta \neq 0. \quad (4)$$

Raw Google and Euclidean coefficients are not directly comparable because the accessibility variables are measured on different scales. I therefore report 2 scale-adjusted mag-

nitudes,

$$\widehat{\text{IQR}}_m = \hat{\beta}_m [P_{75}(A^m) - P_{25}(A^m)], \quad \widehat{\text{SD}}_m = \hat{\beta}_m \cdot \sigma(A^m), \quad m \in \{G, E\}, \quad (5)$$

following Nieminen (2022) and Oka (2021). I convert log-point effects to percentages using $100[\exp(\cdot) - 1]$.

6 Preliminary Results

Table 1 reports a preliminary estimation of equation (2) on the 1-year event window. The Google baseline yields $\hat{\beta}_G = -0.0174$ (SE = 0.0098, $p = 0.077$) with a 95% confidence interval of $[-0.0367, 0.0019]$ that just contains zero, while the Euclidean baseline yields $\hat{\beta}_E = -0.0075$ (SE = 0.0112, $p = 0.502$) and is statistically indistinguishable from zero. Adding the unit-level controls X_{ijt} moves Google to -0.0145 ($p = 0.138$) and Euclidean to -0.0028 ($p = 0.798$). The Google specifications consistently point to a negative gradient at borderline conventional significance, while the Euclidean specifications are centred on zero. A coefficient plot of all 4 specifications with 95% confidence intervals is reproduced in Appendix C as Figure 3.

Table 1. STT Accessibility and Transaction Prices

	(1) Google baseline	(2) Euclidean baseline	(3) Google + controls	(4) Euclidean + controls
Access $\times$ Post	-0.0174* (0.0098)	-0.0075 (0.0112)	-0.0145 (0.0098)	-0.0028 (0.0111)
Observations	16,733	16,733	16,559	16,559
Estates	505	505	500	500
Months	24	24	24	24
Estate FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Unit Controls	No	No	Yes	Yes
First/Second Hand Control	No	No	Yes	Yes
Clustered SE	Estate	Estate	Estate	Estate

Notes: Outcome is log transaction price per square foot. The main sample is restricted to transactions from 2023-03-08 through 2025-03-07. Standard errors are clustered by estate. Google accessibility and Euclidean accessibility follow equation (1). Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Because the two regressors have different scales, magnitude must be assessed after rescaling. Table 2 reports the IQR and standardized-coefficient effects defined in equa-

tion (5). An interquartile move in A^G corresponds to a -0.62% price differential at baseline and -0.53% with controls, against -0.45% and -0.17% for the Euclidean counterparts. The standardized ranking is identical at -0.43% and -0.36% versus -0.26% and -0.10% . Under the main event-window design, route-based accessibility therefore captures a larger and more precise short-run gradient than Euclidean distance.

Table 2. Comparable Effect Sizes of STT Accessibility Measures

	Google baseline	Euclidean baseline	Google + controls	Euclidean + controls
Raw beta	-0.0174	-0.0075	-0.0145	-0.0028
Standard error	0.0098	0.0112	0.0098	0.0111
p-value	0.077	0.502	0.138	0.798
Access P_{25}	-3.0172	-2.6721	-3.0180	-2.6721
Access P_{75}	-2.6614	-2.0751	-2.6485	-2.0751
Access IQR	0.3557	0.5971	0.3694	0.5971
IQR effect, log points	-0.0062	-0.0045	-0.0053	-0.0017
IQR effect, %	-0.62	-0.45	-0.53	-0.17
Standardized beta	-0.0044	-0.0026	-0.0036	-0.0010
Standardized effect, %	-0.43	-0.26	-0.36	-0.10

Notes: Raw Google and Euclidean coefficients are not directly comparable because the accessibility measures use different scales. IQR effects equal $\hat{\beta} \times (P_{75} - P_{25})$, and standardized effects equal $\hat{\beta} \times SD(\text{access})$; percentage effects use $100[\exp(\cdot) - 1]$. Columns correspond to the specifications in Table 1.

I interpret the estimate as the capitalization of the net information released on 8 March 2024. More route-accessible estates experienced weaker relative post-announcement price growth, consistent with short-run local exposure costs from resumption, brownfield reorganization, and construction disturbance outweighing expected long-run accessibility gains within the first year. The result identifies a negative route-based gradient in the announcement window; it does not show that STT reduced prices in a causal counterfactual.

7 Diagnostics and Spatial Pattern

I use 5 diagnostics to probe the leading threats; full tables and regression specifications are reported in Appendix C. Adding a full set of web-scope-by-post interactions absorbs area-specific post trends and yields $\hat{\beta}_G = -0.0453$ ($p = 0.002$) and $\hat{\beta}_E = -0.0728$ ($p =$

0.001). Within web-scope areas, both metrics detect a negative gradient, suggesting that the Google advantage in the main specification is specific to the cross-area event-window design. Conditioning on transaction type leaves $\hat{\beta}_G = -0.0162$ ($p = 0.095$), providing little evidence that first- versus second-hand composition drives the result. Dropping the 5% and 10% closest estates leaves $\hat{\beta}_G = -0.0162$ and -0.0151 , suggesting that the gradient is not driven by a small set of near-STT outliers. Estimating on the full 31,409-transaction sample returns $\hat{\beta}_G = -0.0168$ ($p = 0.087$) and $\hat{\beta}_E = -0.0074$ ($p = 0.507$), so the preliminary result is not confined to the 1-year event window. Finally, an alternative shock-date check replacing the gazettal with the 20 September 2024 approval date shrinks $\hat{\beta}_G$ to -0.0026 ($p = 0.646$) and $\hat{\beta}_E$ to essentially zero, consistent with the response being tied to the gazettal rather than the later administrative confirmation.

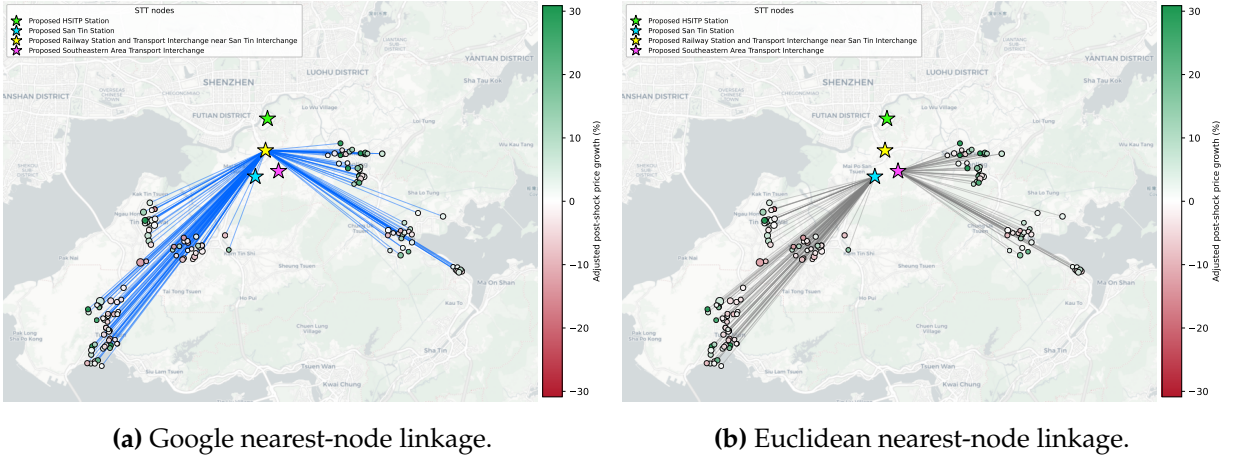


Figure 1. Estate-level adjusted post-pre price growth for estates with at least 10 pre and 10 post transactions. Panels show Google and Euclidean nearest-node linkages. Green or red shading denotes positive or negative residual growth, marker size scales with the square root of transaction count, and star markers denote the 4 STT internal interchanges. The growth measure is the change in estate-level mean hedonic residual log price per square foot between the post and pre windows; its exact construction is reported in equation (7) in Appendix B.

Figure 1 is more informative about spatial clustering than about the accessibility gradient itself. A monotone pattern with distance is not visually obvious, which is why I evaluate the continuous-treatment relationship in the regressions rather than read it directly from the map. What is visually clearer is that nearby estates within the same local cluster

often share similar colours. This matches the diagnostic evidence above: once I add web-scope-by-post interactions, area-specific post-period movements appear important, and both accessibility metrics recover a negative within-area gradient. The map is therefore useful as descriptive context for spatial heterogeneity, while the regressions identify the average gradient of interest. In this sample, the Google-nearest STT node differs from the Euclidean-nearest STT node for all 150 estates, showing that the two metrics assign estates to STT in materially different ways in this part of northern Hong Kong. Heatmaps under the alternative $n \geq 6$ and $n \geq 20$ thresholds are reported in Appendix D.

8 Interpretation and Implications

I interpret the preliminary results as evidence of a short-run negative route-based gradient after the STT gazettal. Within the first year, more route-accessible estates experienced weaker relative price growth, and this pattern is detected more clearly with Google driving time than with Euclidean distance. Across specifications, the Google measure implies larger scale-adjusted effects than the Euclidean measure: IQR effects range from about 0.5% to 0.6% for A^G , versus about 0.2% to 0.5% for A^E . The pattern is consistent with markets pricing short-run local implementation costs from resumption, brownfield reorganization, and construction disturbance ahead of expected long-run accessibility gains during the announcement window; it does not imply that every near-STT estate immediately capitalized the project’s eventual benefits.

The result has three implications. First, mega-project announcements that bundle long-run benefits with short-run local costs should not be modelled as pure amenity shocks: the initial market response may reflect the net of both forces. Second, the choice of accessibility metric is substantively important. Holding the data and specification fixed, Google driving time reveals a borderline-significant negative gradient while the Euclidean coefficient is statistically indistinguishable from zero, consistent with the measurement concerns raised by Apparicio et al. (2008) and Buczkowska et al. (2019). Third,

the timing evidence points to the 8 March 2024 gazettal, rather than the later 20 September 2024 approval, as the relevant information event. For future work on Hong Kong statutory planning, this suggests that the first formal public release may matter more than later administrative confirmation when dating market responses.

9 Limitations and Future Plans

The preliminary analysis has three limitations. First, the time span is short on both sides of the event: the Centanet website’s rolling 3-year public window limits the available pre-period, while the recency of the 2024 gazettal leaves only a short observable post-period. Second, Google driving times are queried in 2025 and applied to transactions from 2023 through 2025. I expect any resulting measurement error to be modest because the local road network changed little over this short period; if present, classical measurement error would attenuate $\hat{\beta}_G$ and bias against finding a Google effect. Third, the estimates identify short-run capitalization gradients rather than long-run effects, and the sign may change as I&T employment, the Northern Link, and construction move from announcement to delivery.

In the full project, I will extend the design in three directions. First, I will estimate an event study with monthly leads and lags, using coefficients β_k on $\mathbf{1}\{t - T^* = k\} \cdot A_i^G$ and a joint test of $\beta_{k < -1} = 0$ as a pre-trend diagnostic. Second, I will estimate a heterogeneity specification that interacts D_{ijt}^G with an estate-level brownfield-exposure indicator constructed from the Town Planning Board’s published brownfield inventory, to probe the local-cost channel. Third, I will extend the data beyond the rolling 3-year Centanet window by merging in Land Registry transaction records. The Centaline Centa-City Leading Index will be used as a descriptive market-wide benchmark rather than as a regressor, since a purely time-varying market-wide index would be absorbed by the month fixed effects in equation (2). Together, these extensions will test pre-trends, sharpen the mechanism, and allow the capitalization gradient to be tracked over a longer horizon as STT

moves from announcement to delivery.

Replication

The data pipeline, regressions, figures, and tables in this proposal are reproducible from the repository at <https://github.com/Yutong428/stt-housing-accessibility>. Reproducing the raw scrape depends on Centanet’s continued public availability of its rolling 3-year transaction window. All downstream code is reproducible from the stored inputs.

References

- Apparicio, P., Abdelmajid, M., Riva, M., and Shearmur, R. (2008). Comparing alternative approaches to measuring the geographical accessibility of urban health services: Distance types and aggregation-error issues. *International Journal of Health Geographics*, 7:7.
- Bao, H. X. H. and Mok, D. K. C. (2020). A link between east and west: How the Guangzhou–Shenzhen–Hong Kong express rail link affects property prices in Hong Kong. *International Real Estate Review*, 23(3):367–395.
- Buczowska, S., Coulombel, N., and de Lapparent, M. (2019). A comparison of Euclidean distance, travel times, and network distances in location choice mixture models. *Networks and Spatial Economics*, 19(4):1215–1248.
- Callaway, B., Goodman-Bacon, A., and Sant’Anna, P. H. C. (2024). Difference-in-differences with a continuous treatment. NBER Working Paper 32117, National Bureau of Economic Research.
- Jayantha, W. M., Lam, T. I., and Chong, M. L. (2015). The impact of anticipated transport improvement on property prices: A case study in Hong Kong. *Habitat International*, 49:148–156.
- Nieminen, P. (2022). Application of standardized regression coefficient in meta-analysis. *BioMedInformatics*, 2(3):434–458.
- Oka, M. (2021). Interpreting a standardized and normalized measure of neighborhood socioeconomic status for a better understanding of health differences. *Archives of Public Health*, 79:226.
- Rojas, A. (2024). Train stations’ impact on housing prices: Direct and indirect effects. *Transportation Research Part A: Policy and Practice*, 181:103979.
- Rosen, S. (1974). Hedonic prices and implicit markets: Product differentiation in pure competition. *Journal of Political Economy*, 82(1):34–55.

Appendix

A Data Construction Pipeline

This appendix documents how I construct the analysis sample and the STT destination nodes.

A.1 From raw listings to a target-area transaction panel

I scrape the public Centanet transaction listings and retain each record’s district, web-scope, estate, building, unit, floor, transaction date, saleable area, transaction price per square foot, and first- or second-hand indicator. Because Centanet exposes a rolling 3-year public window, the raw scrape yields the 283,173 records available at the time of collection rather than a fixed historical archive. I then restrict the data to sale transactions with a non-empty estate name and to the 6 STT-adjacent *web_scope* areas listed in Section 4, leaving 34,495 transactions linked to 720 distinct estate-level geocode queries.

A.2 Estate geocoding and STT internal nodes

For each unique estate, I construct a geocoding query in the format

estate_name, [*big_estate_name*,] *admin_district*, Hong Kong,

including the bracketed parent-estate field when Centanet provides one. I send each unique query to the Google Geocoding API once, cache the result, and broadcast the resulting coordinates back to all matching transactions.

I locate the 4 STT internal interchanges in Table 3 by overlaying the Recommended Outline Development Plan shown on the San Tin Technopole project website⁵ (Figure 2) on Google Maps and reading off the WGS84 coordinates. I exclude Lok Ma Chau Station

⁵Northern Metropolis San Tin Technopole, “Revised Recommended Outline Development Plan.”

because it lies inside the cross-border control area, where the Routes API does not return reliable driving routes.

Table 3. Four STT destination nodes used as accessibility anchors.

Node	Latitude	Longitude
Proposed HSITP Station	22.5222	114.0802
Proposed San Tin Station	22.4869	114.0722
Proposed Railway Station near San Tin Interchange	22.5029	114.0790
Proposed Southeastern Area Transport Interchange	22.4903	114.0875

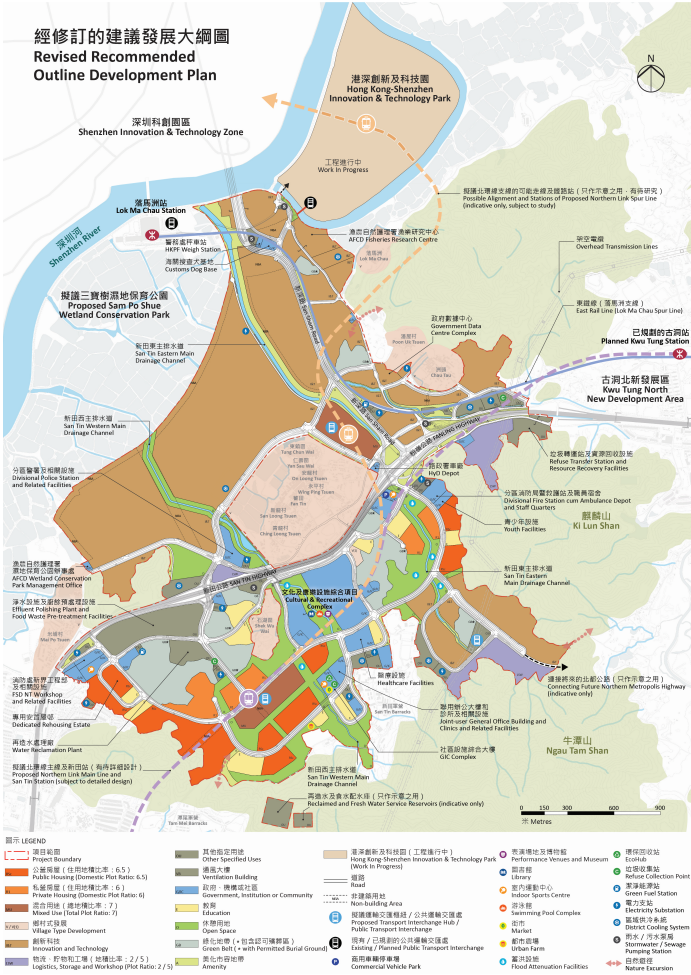


Figure 2. Revised Recommended Outline Development Plan of the San Tin Technopole, used to locate the 4 destination nodes in Table 3.

A.3 Accessibility computation

For each of the 720 unique estate queries and each of the 4 STT nodes, I query the Google Routes API for driving duration and route distance, yielding $720 \times 4 = 2,880$ cached estate-node pairs. I compute the Google and Euclidean accessibility measures as follows.

Algorithm 1 Construct estate-level accessibility measures

Require: Estate coordinates $\{(\phi_i, \lambda_i)\}$, STT nodes $\{(\phi_j, \lambda_j)\}_{j=1}^4$, Google Routes API

- 1: ϕ denotes latitude, λ denotes longitude, and all coordinates are expressed in radians
 - 2: R denotes the Earth's radius in kilometres
 - 3: **for** each estate i **do**
 - 4: **for** each STT node $j \in \{1, \dots, 4\}$ **do**
 - 5: $T_{ij}^G \leftarrow$ Routes API driving duration from estate i to node j in minutes
 - 6: $D_{ij}^E \leftarrow 2R \arcsin\left(\sqrt{\sin^2\left(\frac{\phi_j - \phi_i}{2}\right) + \cos(\phi_i) \cos(\phi_j) \sin^2\left(\frac{\lambda_j - \lambda_i}{2}\right)}\right)$
 - 7: **end for**
 - 8: $T_i^G \leftarrow \min_j T_{ij}^G$
 - 9: $D_i^E \leftarrow \min_j D_{ij}^E$
 - 10: $A_i^G \leftarrow -\log(T_i^G + 1)$
 - 11: $A_i^E \leftarrow -\log(D_i^E + 1)$
 - 12: **end for**
 - 13: Broadcast (A_i^G, A_i^E) to every transaction belonging to estate i
-

B Residual Growth Used in the Heatmap

The colour scale in Figure 1 is not the raw post-pre change in log price per square foot, since that would confound the gradient of interest with changes in observable unit composition between the pre and post windows. I therefore construct an adjusted post-pre growth measure from a hedonic regression that conditions on observable unit attributes but omits estate fixed effects and STT accessibility, so that estate-specific deviations re-

main in the residual. I first estimate

$$\begin{aligned} \log P_{ijt} = & \theta_0 + \theta_1 \log(\text{area}_{ijt}) + \theta_2 \text{age}_{ijt} \\ & + \sum_f \theta_{3f} \mathbf{1}\{\text{floor_group}_{ijt} = f\} + \sum_q \theta_{4q} \mathbf{1}\{t = q\} + u_{ijt}. \end{aligned} \quad (6)$$

on transactions in the 1-year window. The residual

$$\hat{u}_{ijt} = \log P_{ijt} - \widehat{\log P}_{ijt}$$

measures the price of transaction j relative to a unit with comparable observable attributes in the same calendar month. I then collapse the residuals to the estate level,

$$\widehat{\text{Growth}}_i = \frac{1}{|J_i^{\text{post}}|} \sum_{(j,t) \in J_i^{\text{post}}} \hat{u}_{ijt} - \frac{1}{|J_i^{\text{pre}}|} \sum_{(j,t) \in J_i^{\text{pre}}} \hat{u}_{ijt}, \quad (7)$$

where J_i^{pre} and J_i^{post} collect the transactions of estate i before and after the gazettal within the 1-year window. The map shades estate i green if $\widehat{\text{Growth}}_i > 0$ and red otherwise, and scales the marker as

$$\sqrt{|J_i^{\text{pre}}| + |J_i^{\text{post}}|}$$

so that estates with more transactions are visually more prominent without dominating the display.

C Diagnostic Specifications and Tables

This appendix reports the 5 diagnostic checks discussed in Section 7. Each check modifies the baseline design in one dimension to probe a specific explanation for the negative Google gradient. Figure 3 plots the 4 main coefficients with 95% confidence intervals.

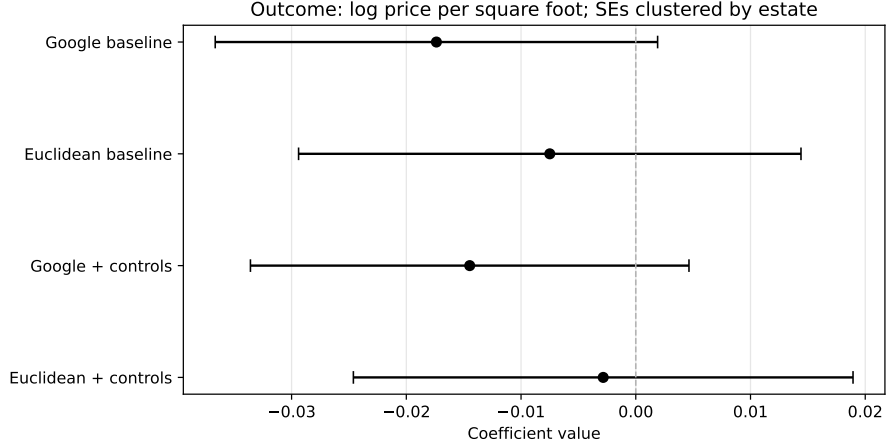


Figure 3. Point estimates and 95% confidence intervals for the 4 main specifications. Standard errors are clustered by estate.

C.1 Web-scope-by-post controls

This check tests whether the main gradient is driven by area-specific post-period movements. I augment the baseline design with a full set of web-scope-by-post interactions,

$$Y_{ijt} = \beta (A_i^m \cdot \text{post}_{ijt}) + \sum_s \eta_s (\mathbf{1}\{\text{web_scope}_i = s\} \cdot \text{post}_{ijt}) + X'_{ijt} \gamma + \alpha_i + \delta_t + \varepsilon_{ijt}, \quad m \in \{G, E\}. \quad (8)$$

The control vector is omitted in the baseline columns. Identification of β then comes from within-web-scope accessibility variation. Results are reported in Table 4.

Table 4. Diagnostic Check 1: Web-Scope Post Controls

	Google baseline	Euclidean baseline	Google + controls	Euclidean + controls
Access $\times$ Post	-0.0453*** (0.0149)	-0.0728*** (0.0216)	-0.0469*** (0.0148)	-0.0752*** (0.0213)
Observations	31,409	31,409	30,308	30,308
Estates	560	560	554	554
Estate FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Web scope $\times$ Post	Yes	Yes	Yes	Yes
First/Second Hand Control	No	No	No	No
Unit Controls	No	No	Yes	Yes
Clustered SE	Estate	Estate	Estate	Estate

Notes: Web scope by post-period controls are included. Post is defined as transaction date on or after 2024-03-08. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 First- versus second-hand controls

This check tests whether the gradient reflects a shift in the mix of primary- and secondary-market transactions near STT. I add a first- versus second-hand indicator to the baseline design,

$$Y_{ijt} = \beta (A_i^m \cdot \text{post}_{ijt}) + X'_{ijt}\gamma + \zeta_{h(i,j,t)} + \alpha_i + \delta_t + \varepsilon_{ijt}, \quad m \in \{G, E\}. \quad (9)$$

Results are reported in Table 5.

Table 5. Diagnostic Check 2: First-Hand/Second-Hand Controls

	Google baseline	Euclidean baseline	Google + controls	Euclidean + controls
Access $\times$ Post	-0.0162* (0.0097)	-0.0054 (0.0111)	-0.0163* (0.0097)	-0.0046 (0.0111)
Observations	31,409	31,409	30,308	30,308
Estates	560	560	554	554
Estate FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Web scope $\times$ Post	No	No	No	No
First/Second Hand Control	Yes	Yes	Yes	Yes
Unit Controls	No	No	Yes	Yes
Clustered SE	Estate	Estate	Estate	Estate

Notes: First-hand/second-hand controls are included. Post is defined as transaction date on or after 2024-03-08. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Dropping the nearest estates

This check tests whether the Google gradient is driven by a small set of estates closest to STT. I re-estimate the Google specifications after excluding estates in the top 5% and top 10% of the estate-level Google accessibility distribution, retaining the subsamples

$$\{i : A_i^G < P_{95}(A^G)\} \quad \text{and} \quad \{i : A_i^G < P_{90}(A^G)\}.$$

Results are reported in Table 6.

Table 6. Diagnostic Check 3: Dropping Estates Nearest to STT

	Original Google baseline	Original Google + controls	Drop top 5% baseline	Drop top 5% + controls	Drop top 10% baseline	Drop top 10% + controls
Access $\times$ Post	-0.0168* (0.0098)	-0.0174* (0.0098)	-0.0162 (0.0101)	-0.0167* (0.0101)	-0.0151 (0.0104)	-0.0156 (0.0104)
Observations	31,409	30,308	31,071	29,970	30,713	29,612
Estates	560	554	532	526	502	496
Estate FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Web scope $\times$ Post	No	No	No	No	No	No
First/Second Hand Control	No	No	No	No	No	No
Unit Controls	No	Yes	No	Yes	No	Yes
Clustered SE	Estate	Estate	Estate	Estate	Estate	Estate

Notes: Drop samples exclude estates above the estate-level Google-accessibility percentile cutoff. Post is defined as transaction date on or after 2024-03-08. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.4 Full sample

This check tests whether the main result is confined to the symmetric 1-year event window. I re-estimate the 4 main specifications on the full Centanet analysis sample. Results are reported in Table 7.

Table 7. Diagnostic Check 4: Full Sample

	Google baseline	Euclidean baseline	Google + controls	Euclidean + controls
Access $\times$ Post	-0.0168* (0.0098)	-0.0074 (0.0111)	-0.0174* (0.0098)	-0.0075 (0.0111)
Observations	31,409	31,409	30,308	30,308
Estates	560	560	554	554
Estate FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Web scope $\times$ Post	No	No	No	No
First/Second Hand Control	No	No	No	No
Unit Controls	No	No	Yes	Yes
Clustered SE	Estate	Estate	Estate	Estate

Notes: The sample uses all available observations. Post is defined as transaction date on or after 2024-03-08. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.5 Alternative shock date at approval

This check tests whether the result depends on dating the shock at the 8 March 2024 draft-plan gazettal. I replace the main post indicator with one based on the 20 September 2024 approved-OZP publication date,

$$Y_{ijt} = \beta (A_i^m \cdot \text{post}_{ijt}^{\text{appr}}) + X'_{ijt}\gamma + \alpha_i + \delta_t + \varepsilon_{ijt}, \quad m \in \{G, E\}, \quad (10)$$

$$\text{post}_{ijt}^{\text{appr}} = \mathbf{1}\{\tau_{ijt} \geq 2024-09-20\}.$$

Results are reported in Table 8.

Table 8. Diagnostic Check 5: Approval Shock Date

	Google baseline	Euclidean baseline	Google + controls	Euclidean + controls
Access $\times$ Post	-0.0026 (0.0056)	-0.0001 (0.0070)	-0.0025 (0.0057)	-0.0001 (0.0072)
Observations	31,409	31,409	30,308	30,308
Estates	560	560	554	554
Estate FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Web scope $\times$ Post	No	No	No	No
First/Second Hand Control	No	No	No	No
Unit Controls	No	No	Yes	Yes
Clustered SE	Estate	Estate	Estate	Estate

Notes: Post is defined as transaction date on or after 2024-09-20, the approved San Tin Technopole OZP publication date. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Additional Heatmaps

Figures 4 and 5 reproduce the spatial heatmaps in Figure 1 under the alternative $n \geq 6$ and $n \geq 20$ inclusion thresholds. The qualitative pattern is preserved as the threshold becomes more or less restrictive.

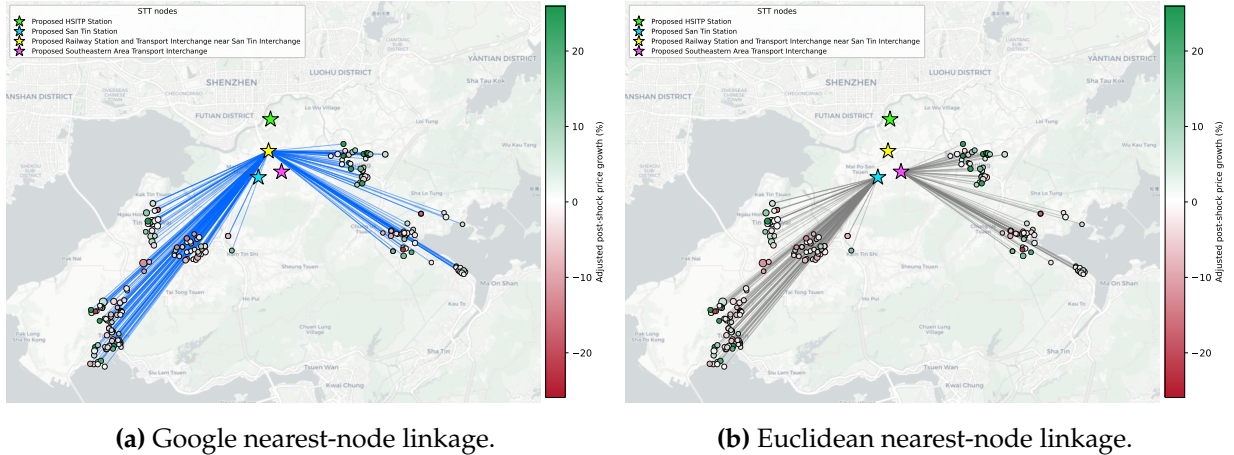
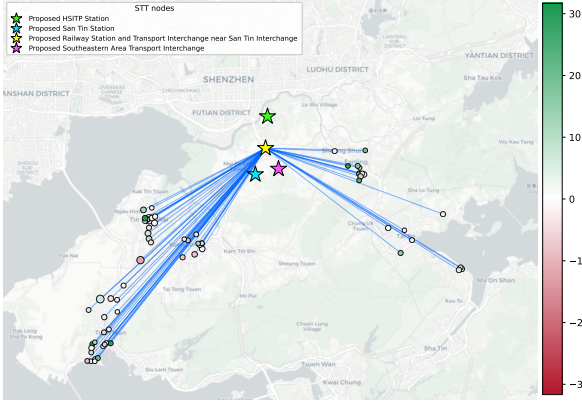
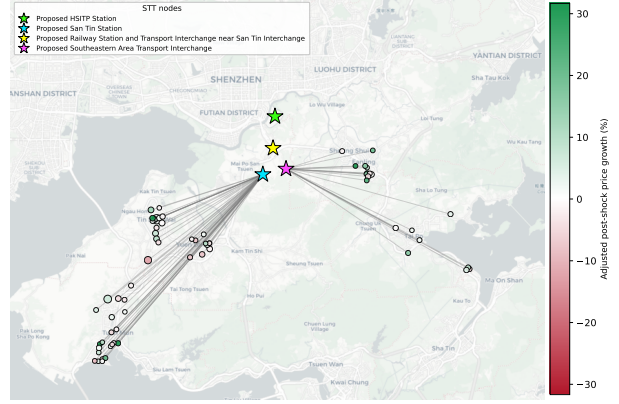


Figure 4. Estate-level adjusted post-pre price growth for estates with at least 6 pre and 6 post transactions. Panels show Google and Euclidean nearest-node linkages. Green or red shading denotes positive or negative residual growth, marker size scales with the square root of transaction count, and star markers denote the 4 STT internal interchanges. The growth measure is the change in estate-level mean hedonic residual log price per square foot between the post and pre windows; its exact construction is reported in equation (7) in Appendix B.



(a) Google nearest-node linkage.



(b) Euclidean nearest-node linkage.

Figure 5. Estate-level adjusted post-pre price growth for estates with at least 20 pre and 20 post transactions. Panels show Google and Euclidean nearest-node linkages. Green or red shading denotes positive or negative residual growth, marker size scales with the square root of transaction count, and star markers denote the 4 STT internal interchanges. The growth measure is the change in estate-level mean hedonic residual log price per square foot between the post and pre windows; its exact construction is reported in equation (7) in Appendix B.